



IARE
INSTITUTE OF
AERONAUTICAL ENGINEERING



SAE INTERNATIONAL - AERO DESIGN WEST 2025

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Team Number: 21431953

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APPENDIX A – STATEMENT OF COMPLIANCE

Certificate of Qualification

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Statement of Compliance

As Faculty Advisor:

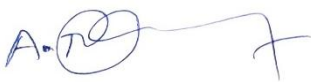
Dr. A. Rathan Babu, I certify that the registered team members are enrolled in collegiate courses.

Dr. A. Rathan Babu, I certify that this team has designed and constructed the radio – controlled aircraft in the past nine (9) months with the intention to use this aircraft in 2025 SAE Aero Design competition, without direct assistance from professional engineers, R/C model experts, and/or related professionals.

Dr. A. Rathan Babu, I certify that this year's Design Report has original content written by members of this year's team.

Dr. A. Rathan Babu, I certify that all reused content has been referenced and is in compliance with the University's plagiarism and reuse policies.

Dr. A. Rathan Babu, I certify that the team shall reuse the Requirements Check and Safety and Airworthiness Inspection checklists to inspect their aircraft before arrival at Technical Inspection and that the team shall submit the completed checklists, signed by the Faculty Advisor or Team Captain, to the inspectors before Technical Inspection begins.



Signature of Faculty Advisor

Date: 2/16/2025



Signature of Team Captain

Date: 2/16/2025

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1 INTRODUCTION

1.1 Background

The Aero Design 2025 competition provides undergraduate and graduate engineering students a challenging environment for aircraft development. It addresses several real-life engineering processes from initial conceptual designs to final manufacturing. It also covers all system integration and the entirety of mission performance. This UAV design offers a novel approach. It fulfills the competition's technical and functional requirements. This project will build a useful, reliable and ready-to-use drone using the latest technology in aerodynamics, structure and propulsion.

1.2 Design Objective

To guarantee efficient flight, the unmanned aerial vehicle needs to minimize drag and maximize lift while it carries its loads. The aircraft maintaining high degree of stability and offer highly responsive maneuverability during all phases of flight take-off, cruising and landing; to guarantee reliable operation. Designing that the fuselage design allows for the smooth integration of liquid payloads while upholding its structural integrity under all running loads and condition.

1.3 Design Constraints

Table 1: Design Constraints

S. No	Design Feature	Requirement
1	Aircraft Type	Fixed Wing UAV
2	Propulsion System	All Electric
3	Gross Take-Off Weight	55 Pounds
4	Maximum Battery	Max 4 Cells
5	Payload	67 Fluid Ounces
6	Each Aircraft Component Dimension	< 48 Inches

1.4 Mission Constraints

Table 2: Mission Constraints

S. No	Mission Feature	Requirement
1	Take-Off Distance	100ft
2	Initial Turn Distance	400ft
3	Landing Distance Limit	200ft
4	Transmitter	2.4 Hz
5	Number of Attempts	1

1.5 Mission Profile and Work Flow

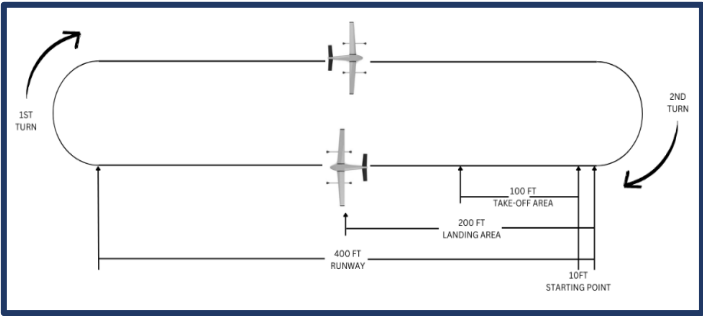


Figure 1: Mission Profile

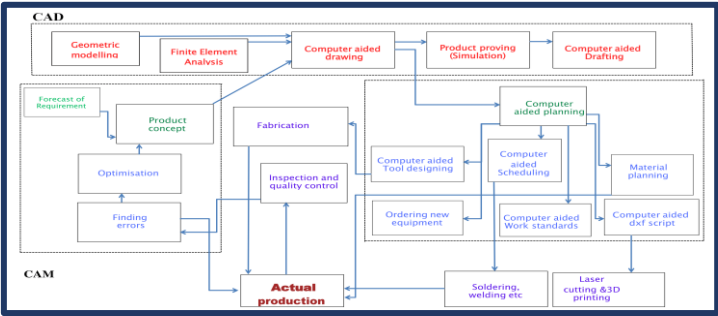


Figure 2: Work Flow

2 DESIGN EVALUATION

2.1 Literature Survey

- **CFD investigation of airfoil shaped fuselage-** The research paper examines an airfoil-shaped fuselage and its aerodynamic characteristics at varying velocities and angles of attack. CFD analysis shows that the fuselage shape increases lift.
- **Aerodynamic optimization of high-wing aircraft-** The study optimizes fuselage-wing shape in three stages: refining the fuselage fairing, modifying the nose for lift, and improving the lift-to-drag ratio. This design achieves 1.5x higher lift coefficient than the DLR-F6 benchmark.
- **Landing gear layout design for unmanned aerial vehicle-** This research focuses on designing stable, shock-absorbent drone landing gear with a tricycle setup.

- **Stability analysis of a sprayer UAV with a liquid tank-** This study examines how liquid sloshing affects UAV by analyzing tank shapes and internal structures. Results show that a hexagonal tank with baffles minimize turbulence and improved flight stability
- **Fabrication of fixed-wing UAV** - A strong, efficient fixed-wing drone using materials like carbon fiber and balsa wood, combining CNC cutting, 3D printing, and manual craftsmanship.
- **Essential parts for an UAV-** Drones consist of key elements like wings for lift, a fuselage for payload, a tail for stability, and landing gear for smooth take-off and landing.
- **Sloshing inside payload container-** Sloshing affects stability due to liquid movement in partially filled tanks, controlled using baffles and reinforced structures, using a cuboid container for maximum capacity, high-strength 3D-printed material for durability, and baffles to stabilize water flow and maintain aircraft stability.

2.2 Wing Planform

Table 3: Wing Profile Geometry

Type	Figure of Merit	L/D	Stability	Structural Weight	Ease of Manufacture	Total
	Score Factor	0.4	0.2	0.3	0.1	Score (1)
Wing Planform	Rectangle	0	0	0	1	0.1
	Tapered	1	1	1	0	0.9
	Swept	1	1	0	0	0.6
Wing Position	High Wing	1	1	1	1	1
	Mid Wing	0	1	1	0	0.5
	Low Wing	0	0	1	1	0.4

This high-wing configuration is tapered and allows a decrease of induced drag and a good lift distribution to broaden the range of aerodynamic efficiencies. Accordingly, giving good stability and visibility and adapted to work and a simpler attachment of the payload.

2.3 Fuselage Planform

Table 4: Fuselage Profile Geometry

Type	Figure of Merit	Efficiency	Payload Distribution	Ease of Manufacture	Total Score
	Score Factor	0.4	0.3	0.3	(1)
Fuselage Planform	Circular	1	0	1	0.7
	Rectangular	1	0	1	0.7
	Cambered Airfoil Cross Section	1	1	1	1
	Elliptical	1	0	0	0.4

It has a cambered airfoil cross-section of fuselage to minimize drag and enhance lift, improving overall performance efficiency. The flat bottom allows landing gear and payload bay to have functional compatibility and ease of mounting.

2.4 Tail and Landing Gear Planform

Table 5: Tail Profile Geometry

Type	Figure of Merit	Drag	Stability	Structural Weight	Ease of Manufacture	Total
	Score Factor	0.4	0.3	0.2	0.1	Score (1)
Tail Planform	Conventional Tail	0	1	1	1	0.6
	V – Tail	0	1	0	0	0.2
	H – Tail	0	1	0	1	0.4
	T – Tail	1	0	0	0	0.3

A standard configuration has been selected for ease of engineering and to reduce the effects of the payload's slosh-induced instability in flight.

Table 6: Landing Gear Profile Geometry

Type	Figure of Merit	Weight	Power	Efficiency	Thrust Factor	Total Score (1)
	Score Factor	0.2	0.3	0.3	0.2	
Landing Gear Configuration	Pusher Tricycle	1	1	0	0	0.5
	Puller Tricycle	1	1	1	1	1
	Tail Dragger	0	1	0	1	0.5

The tricycle undercarriage, using a puller arrangement, improves aerodynamic perfection, lowers the take-off distance, and enhances ground stability.

2.5 Overall Configurations

Table 7: Overall Configurations

Part	Type
Wing	Tapered, High Wing
Fuselage	Cambered Airfoil Cross-Section
Tail	Conventional
Landing Gear	Puller Tricycle

3 AIRFOIL ANALYSIS

3.1 Airfoil Selection

Table 8: Airfoil Selection & Interpolation

Type	Figure of Merit	C_L	C_m VS α	C_L/C_D	Ease of Manufacture	Total Score (1)
	Score Factor	0.3	0.2	0.2	0.3	
Airfoil Selection	NACA 2412	1	0	1	0	0.5
	NACA 4412	1	1	0	0	0.5

NACA 6412	0	1	0	0	0.2
CLARK Y	1	0	1	1	0.8
RAF 32	1	1	1	0	0.7
NACA 2412 + NACA 4412 – (Interpolated)	1	0	0	0	0.3
NACA 6412 + CLARK Y – (Interpolated)	1	1	0	0	0.5
(NACA 2412 + NACA 4412) + (NACA 6412 + CLARK Y) – (Interpolated)	1	1	1	1	1

The flat-bottom airfoils were chosen because of their superior lift. Refinement of this design led to an optimized airfoil profile that fulfilled the operation requirements. Improved aerodynamics results from the shape manipulation with increased lift over that of previously used designs with gains in the overall wing efficiency, stability, and performance.

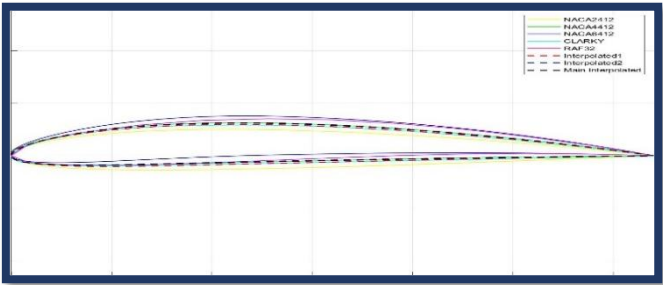


Figure 3: Airfoil

3.2 Airfoil Analysis

The analysis for the pressure distribution was carried out in the ANSYS software. The K-Omega SST model is chosen as it is the most accurate model for subsonic analysis as per ANSYS guide book. The inlet velocity for domain is 10 m/s. The total number of nodes of the mesh are 70834, and total number of elements are 70166. Various sizing for airfoil

component and domain is applied, including the edges and face, for a refined mesh, so the results obtained would produce better accuracy and precision. The velocities are divided into components, horizontal and vertical components, so that the direction of the resultant velocity component changes which results in the change of the angle of attack.

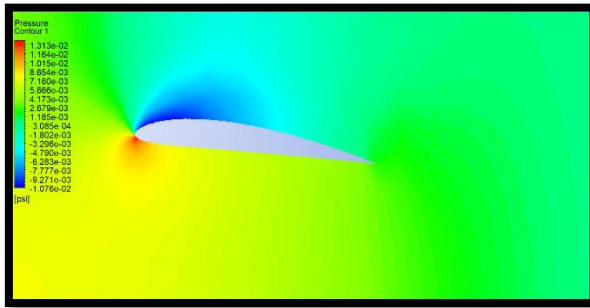


Figure 4: Angle of Attack 7

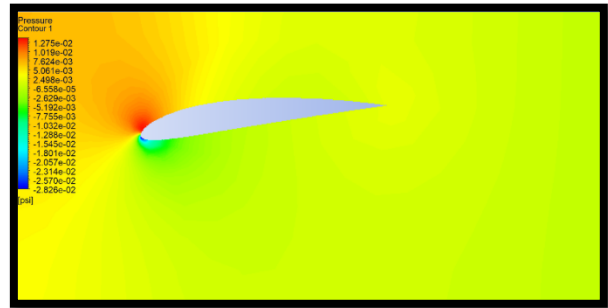


Figure 5: Angle of Attack -7

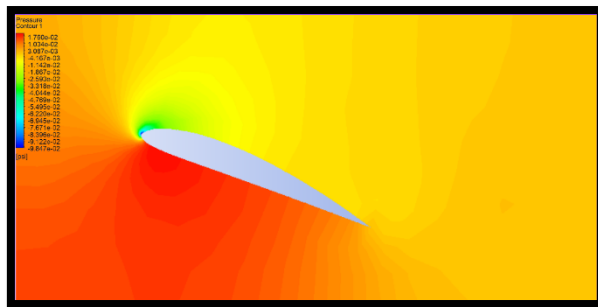


Figure 6: Angle of Attack 15

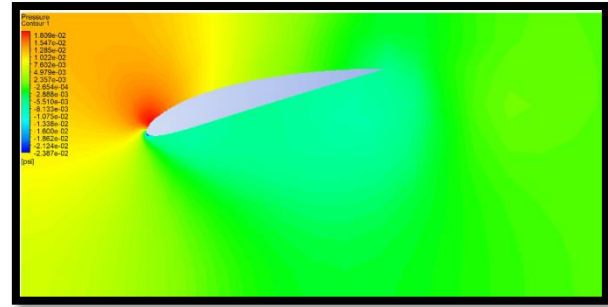


Figure 7: Angle of Attack -15

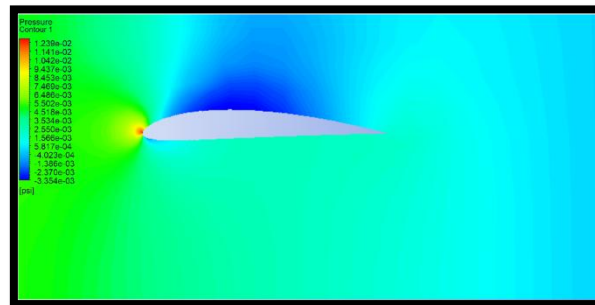


Figure 8: Angle of Attack 0

4 POWER PLANT SELECTION

4.1 Motor Selection

The T-Motor AS2820 Long Shaft 1250KV (Table 9) is ideal for fixed-wing UAVs, offering high efficiency, robust performance, and a lightweight design with strong thrust.

Table 9: Motor Selection

Motor	AT2321 Long Shaft	AS2820 Long Shaft	AT2820 Long Shaft
			
KV	950	1250	1050
Weight(pounds)	0.2050	0.3175	0.3175
Rated Voltage (LiPo)	2-4s	3-4S	3-4S
Peak Current	30	77A	67A
Idle Current	1.2A	3.44A	2.59A
Thrust at 75% Throttle	2.4757	4.3916	4.7509

4.2 Avionics Selection (ESC & Battery)

The Hobbywing Skywalker 80A ESC (Table 10) was chosen for its motor compatibility, and tested suitability.

Table 10: ESC Selection

ESC	Hobbywing Skywalker 60a	Hobbywing Skywalker 80a	Hobbywing Platinum 80
			
Continuous Current(A)	60A	80A	80A
Peak Current(A)	80A	100A	120A
Weight(pounds)	0.1389	0.1808	0.1708
Voltage (LiPo)	2-6S	2-6S	3-8S
Size(mm)	77*35*14	86*38*12	60.5*33*17.6

The Bonka 2200mAh (Table 11) battery was chosen for its lightweight design, 35C continuous discharge, and 70C burst, ensuring compatibility with the motor and ESC.

Table 11: Battery Selection

BATTERY	Turnigy 2200mAh	Bonka 2200mah	Tattu 2200mAh
			
Capacity(mAh)	2200	2200	2200
Voltage(V)	11.1V(3S)	11.1V(3S)	11.1V(3S)
Discharge(C)	25	35	45
Weight(pounds)	0.3924	0.3858	0.4189
Size(mm)	105*35*24	105*34*25	107*36*24

4.3 Propeller Selection

The propeller 12*6E is chosen as it's the most efficient for its performance with T-Motor AS2820 Long Shaft 1250KV, 10*5.5 propeller doesn't withstand the thrust load at higher throttle and 13*6 propeller reduces the battery efficiency.

Table 12: Propeller Selection

Propeller			
Size	10*5.5MR	12*6E	13*6E
Diameter	10 in	12 in	13
Pitch	5.5 in	6 in	6 in
Weight	0.0331lbs.	0.0593 lbs.	0.1056 lbs.
HUB Diameter	0.65 in	0.80 in	1.00 in
HUB Thickness	0.35 in	0.41inch	0.46 in

5 POWER PLANT PERFORMANCE

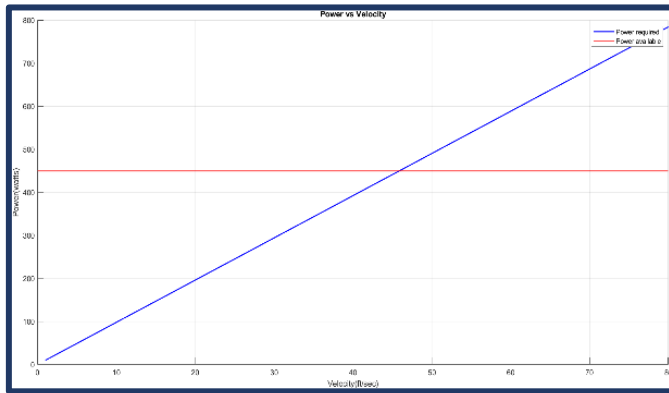


Figure 9: Power vs Velocity

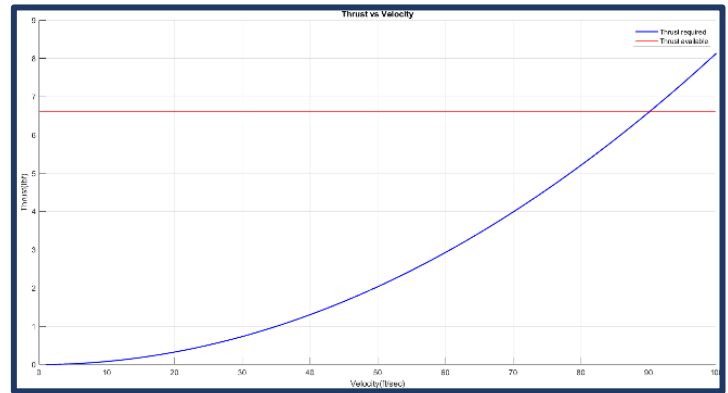


Figure 10: Thrust vs Velocity

The graph (figure 9) shows the relationship between power required and power available for an aircraft. The point where these lines intersect marks the maximum velocity for level flight.

The graph (figure 10) represents the thrust required and the thrust available the point of intersection marks the minimum velocity at which the aircraft can maintain a level flight.

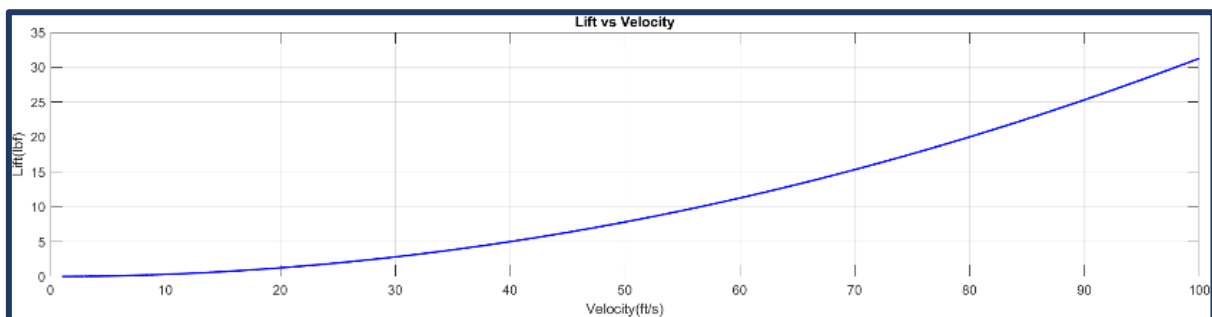


Figure 11: Lift vs Velocity

The curve exhibits a nonlinear increase in lift with the square of velocity the steady upward trend confirms that the wing design effectively produces lift as velocity increase ensuring sufficient lift generation.

6 INITIAL SIZING

6.1 Wing Sizing

The wing is the major component of an airplane responsible for producing a large part of the total lift necessary during the period desired for sustained flight.

Table 13: Wing Data

Parameter	Magnitude (inches)
Span	62.9916 inches
Surface Area	657.2013 inches ²
Root Chord	12.5984 inches
Tip Chord	8.2677 inches
Taper Ratio	0.6562
Aspect Ratio	6.0377
Mean Aerodynamic Chord Length	10.5814 inches
Mean Aerodynamic Chord (from root chord)	14.6685 inches
Aerodynamic Centre	4.6625 inches

6.2 Fuselage Sizing

The fuselage is the core of the aircraft, housing the payload, avionics, and supporting systems. A cambered airfoil cross-section was chosen for optimal aerodynamic performance and a good fit for the payload bay. The flat bottom design simplifies the installation and removal of the landing gear, improving operational ease.

Table 14: Fuselage Sectional Area Breakdown

Fuselage Section	Dimensions(inches)
Total Length of Fuselage	41.2598 inches
Total Length of Fuselage (including) empennage	49.1338 inches
Fuselage Section 1 (Nose Cone)	9.7637 inches
Fuselage Section 2	12.5984 inches
Fuselage Section 3	18.8976 inches
Cross Sectional Area of 1 st Section	31.6913 inches ²
Cross Sectional Area of 2 nd Section	32.6383 inches ²
Cross Sectional Area of 3 rd Section	31.7508 inches ²

6.3 Tail Sizing

Horizontal stabilizers adjust for the pitching moments of an aircraft while tilting upward and downward which results in maintaining longitudinal static stability. The vertical tail limits yawing moments during turns, which in turn allows for lateral movement to the left or right while maintaining directional stability.

$$S_H = \frac{\bar{c} \times S \times V_{HT}}{L_{HT}} \quad S_{VT} = \frac{b \times S \times V_{VT}}{L_{VT}}$$

$L_{HT/VT}$ = Distance from center of gravity of aircraft to aerodynamic center of the Horizontal/Vertical tail

C_{HT} = Volume coefficient horizontal tail = 0.5345; C_{VT} = Volume coefficient for vertical tail = 0.0507

$\bar{c}$ = Mean aerodynamic chord; $S_{HT/VT}$ = Surface area for Horizontal/Vertical tail

Table 15: Tail Data

Parameter	Horizontal Tail	Vertical Tail
Airfoil	NACA 0015	NACA 0015
Span	19.6850 inches	11.8110 inches
Surface Area	124.0001 inches ²	69.7501 inches ²
Root Chord	7.8740 inches	7.8740 inches
Tip Chord	4.7244 inches	3.9370 inches
Taper ratio	0.6	0.5
Aspect Ratio	3.125	2

6.4 Control Surface Sizing

Control surfaces control the aircraft's attitude and stability. Includes ailerons on the wings for controlling roll, elevators on the horizontal tailplane for controlling pitch, and a rudder on the vertical tailplane yaw.

Table 16: Control Surfaces Data

Control Surfaces	Length	Chord
Aileron	25% length of wing span = 15.7480 inch	15% of root chord of wing = 1.8897 inch
Elevator	100% length of the tail = 19.6850 inch	25% of root chord of tail = 1.9685 inch
Rudder	90% length of the fin = 10.6299 inch	25% of root chord of fin = 1.9685 inch

6.5 Servo Sizing

Servos perform the important function of controlling movements on the surfaces of the airframe by converting electrical signals into mechanical motion to orient the ailerons, elevators, rudder, and secondary surfaces. Their use will lead to precise control and stabilizing of Aerodynamics for stable flight conditions.

$$T = \frac{8.5 \times 10^{-6} \times C^2 \times V^2 \times L \times \sin(S_1) \times \tan(S_2)}{\tan(S_2)}$$

T = Torque (lb ft) C = chord length of the control surface; L = span of the control surface

S_1 = maximum control surface deflection = 25° ; S_2 = maximum servo angle deflection = 160°

Table 17: Control Surfaces Torque

Control Surface	Aileron	Elevator	Rudder
Torque	1.1×10^{-4}	1.5×10^{-4}	8.1×10^{-5}

7 AIRCRAFT WEIGHT ESTIMATION

Table 18: Avionics Weight Estimation

S. No	Avionics	Weight (Pounds)
1	Motor Weight	0.3174 lbs.
2	Motor Battery (2200mAh)	0.3858 lbs.
3	ESC	0.1785 lbs.
4	Power Limiter	0.0440 lbs.
5	Servo Battery (1000mAh)	0.1433 lbs.
6	Connecting Wires-Y	0.0176 lbs.
7	Extenders x 6	$0.0132 \times 6 = 0.0793$ lbs.
8	Servos x 5	$0.0242 \times 5 = 0.1212$ lbs.
9	Landing Gear Servos	0.1168 lbs.
10	Propeller	0.0595 lbs.
11	Receiver	0.0330 lbs.
12	Servo Horns	0.0154 lbs.

Table 19: Structural Weight Estimation

S. No	Component	Weight (Pounds)
1	Landing Gear	1.0207 lbs.
2	Ribs	0.3768 lbs.
3	Bulkheads	0.2735 lbs.
4	Spars	1.3155 lbs.
5	Skin	0.4415 lbs.

$$\text{Take-Off Weight} = W_{\text{structural}} + W_{\text{payload}} + W_{\text{avionics}}$$

$$\text{Take-Off Weight} = 3.4281 \text{ lbs. } (W_{\text{structural}}) + 4.6737 \text{ lbs. } (W_{\text{payload}}) + 1.5118 \text{ lbs. } (W_{\text{avionics}})$$

$$\text{Take-Off Weight} = 9.6136 \text{ lbs.}$$

8 CONCEPTUAL DESIGN

8.1 CAD Design

The aircraft and payload bay have been designed in the CATIA V5 and AUTODESK FUSION 360 software, wherein the appropriate material for different components was applied for analysis purposes. The isometric view, front view, side view and the top view have been displayed in the pictures below for aircraft and side view and isometric view for the payload bay. The payload bay has been placed at the CG location of the aircraft so that the stability is not affected due to the discharge rate of the liquid through payload during flight adhering to the mission requirement.



Figure 12: CAD Model Front View



Figure 13: CAD Model Side View

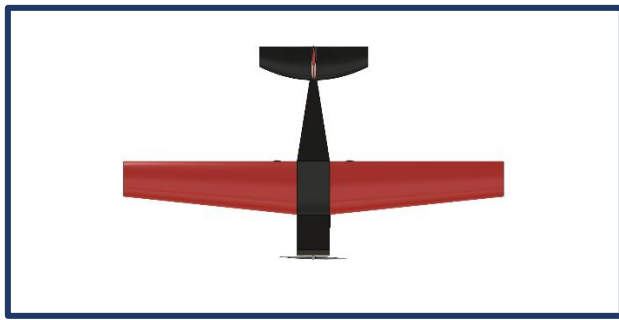


Figure 14: CAD View Top View



Figure 15: CAD Model Isometric View

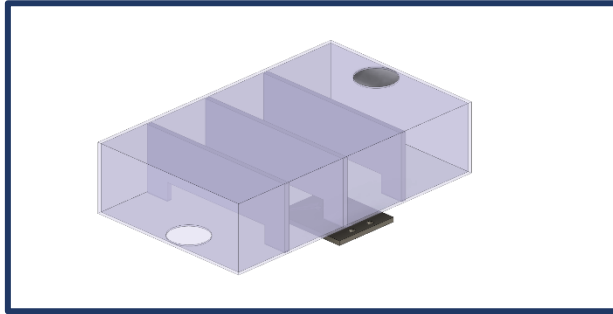


Figure 16: Isometric View of Payload Bay

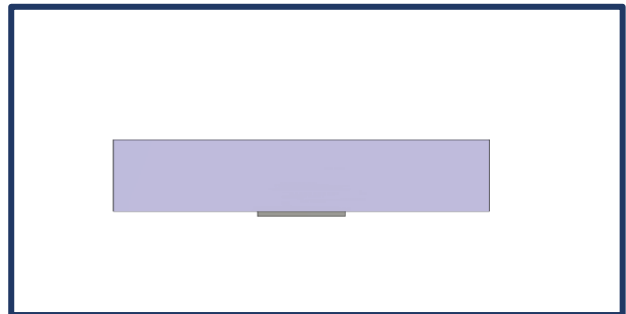


Figure 17: Side View of Payload Bay

9 ANALYSIS

9.1 Structural Analysis of Wing

The structural analysis of the wing has been performed in the Ansys Software, wherein the total number of nodes and elements of the mesh are 103954 and 53523 respectively. Balsa wood has been assigned as the material for the ribs and the I-Sectional beam, and the spars of the wing have been assigned aluminum rods as the material.

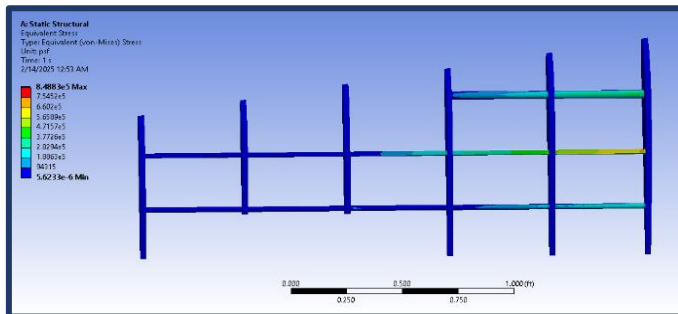


Figure 18: Equivalent Stress of Wing

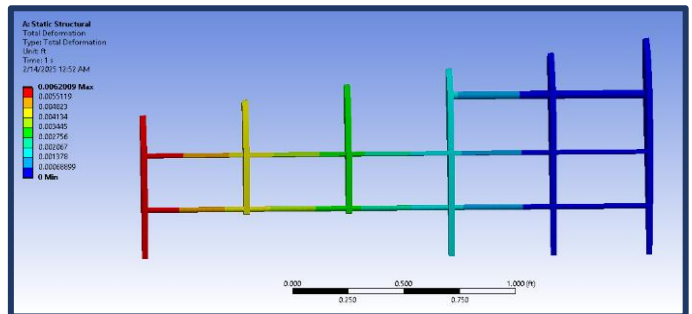


Figure 19: Total Deformation of Wing

Solution	Minimum	Maximum	Average
Total Deformation	0	6.2009e-003	2.051e-003
Equivalent Elastic Strain (ft/ft)	3.8783e-015	2.0445e-004	1.2154e-005
Equivalent Stress(psf)	5.6233e-006	8.4833e+005	49019

9.2 CFD Analysis of Aircraft

The CFD flow analysis of the aircraft was carried out in the ANSYS FLUENT software wherein the aerodynamic characteristics of the aircraft can be thoroughly studied. The contours of velocity and pressure is displayed. The number of the nodes and elements are 276435 and 1532935 respectively. The solution converged at 1231 iterations.

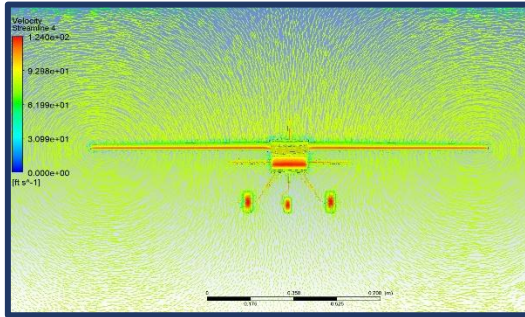


Figure 20: Velocity Contour of Aircraft

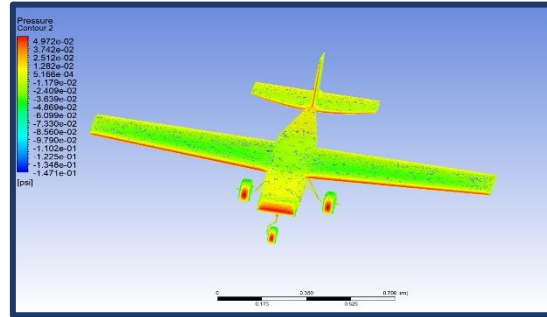


Figure 21: Pressure Contour of Aircraft

9.3 Payload Analysis

The Ansys Fluent is used for the simulation of the payload bay designed. The geometry is first imported and then meshed, and further carried out to the solver section. The multiphase section is selected, wherein the phases 1 and 2 are defined as air and liquid water respectively, with the volume meshing being opened. The surface tension for the water is 0.00518 lb/ft . The volume of fluid (vof) method is chosen, with acceleration being set as 3.5728 ft/s^2 . The incrementations in the calculation section are set to be at 0.1 for upto 1500 time-steps being chosen. The total volume of the 67 fluid ounces is named as water, under the liquid water phase. After the simulation is completed, the result plot files are exported and the MATLAB software is used to plot the graphs.

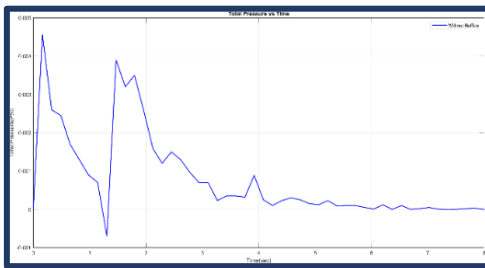


Figure 22: Pressure vs Time

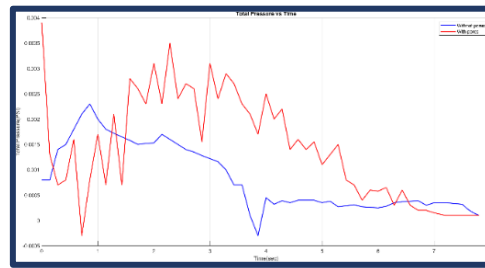


Figure 23: Pressure vs Time 2

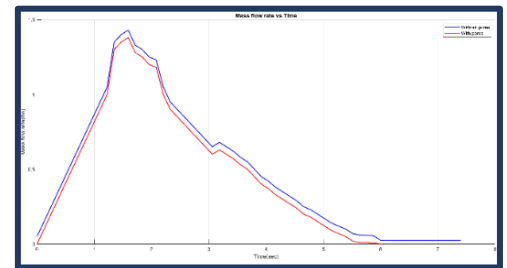


Figure 24: Mass Flow Rate vs Time

- In Figure (22), The force exerted on the outlet area of the container is $4.5\text{e-}3 \text{ PSI}$, wherein the container without the baffles experiences an acceleration nearly for 2.2 seconds, and the movement of the liquid water in the container results in the formation of spikes in the plot. The time taken to attain equilibrium is 6.8 seconds, where the acceleration nears zero.

- In Figure (23), we demonstrate that the baffles without pores perform better than the baffles with pores. The multiple spikes which have been plotted cause the aircraft to experience forward momentum loss of the aircraft which would lead to stalling during take-off. Hence, the use of the baffles without pores is taken into consideration.
- In Figure (24), it is shown that the outlet when completely opened, the container with baffles without pores takes about 6.1 seconds to discharge the liquid water, and the container with baffles with pores takes about 5.7 seconds. Both the types have almost same discharge rate, varying only by a small margin.

9.4 Landing Gear Analysis

The structural analysis of the landing gear has been carried out in the Ansys Software, wherein the total number of nodes and elements of the mesh are 266862 and 148924 respectively. The landing gear has been made of iron rods, the top base has been 3D printed and is attached at the bottom of the fuselage. The wheels are made of pneumatic rubber. The total deformation, equivalent elastic strain, and the equivalent stress of the landing gear have been solved.

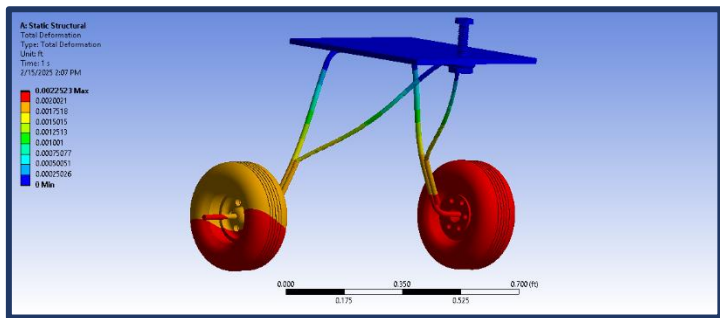


Figure 25: Landing Gear Total Deformation

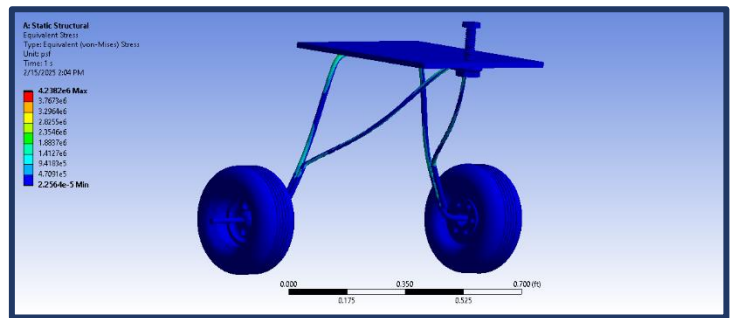


Figure 26: Landing Gear Equivalent Stress

Table 20: Results of Landing Gear

Solution	Minimum	Maximum	Average
Total Deformation (ft)	0	2.2523e-003	1.3908e-003
Equivalent Elastic Strain (ft/ft)	4.412e-014	1.042e-003	6.0025e-006
Equivalent Stress (psf)	2.2564e-005	4.2382e+006	21217

10 STABILITY ANALYSIS

10.1 Static Stability

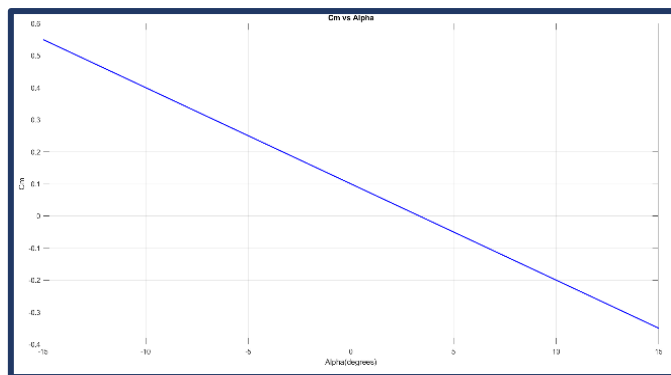


Figure 27: C_m VS Alpha Graph

The graph in figure (27) helps in determining the characteristic properties of the static stability of the aircraft and variation of the pitching moment coefficient C_m with the angle of attack AOA. The pitching moment can be defined as the tendency of the aircraft to pitch along the lateral axis.

10.2 Longitudinal Dynamic Stability

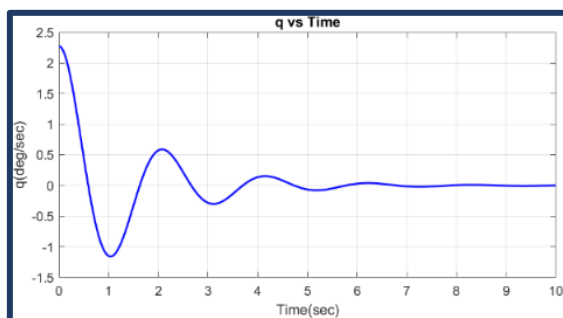


Figure 28: Pitch Rate (q) vs Time

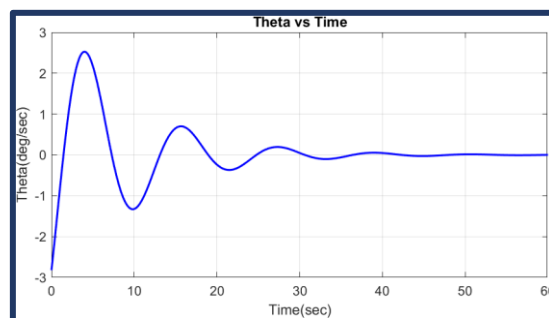


Figure 29: Pitch Angle (θ) vs Time

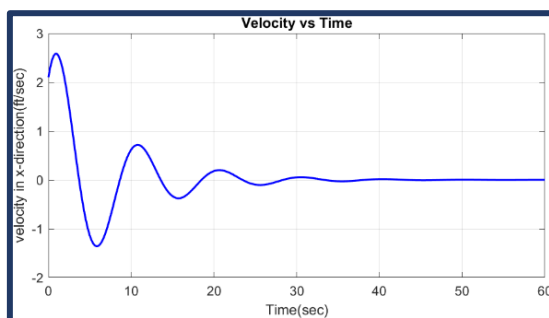


Figure 30: x-Direction Velocity vs Time

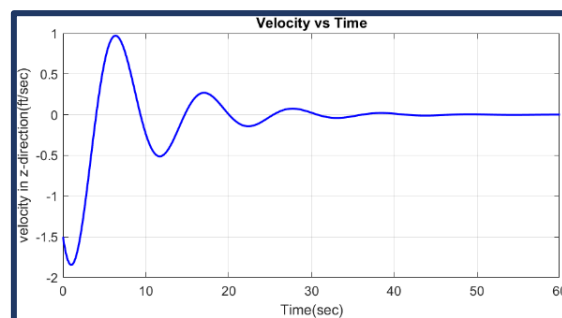


Figure 31: z-Direction Velocity vs Time

The longitudinal dynamic stability refers to the tendency of the aircraft to return to its equilibrium position after being disturbed from its steady state, typically because of various factors such as external loads, turbulence or gusty winds.

10.3 Lateral Dynamic Stability

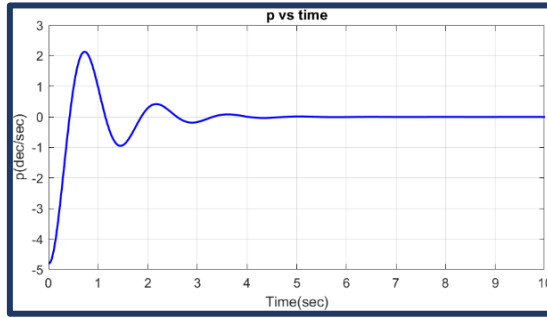


Figure 32: Roll (p) vs Time

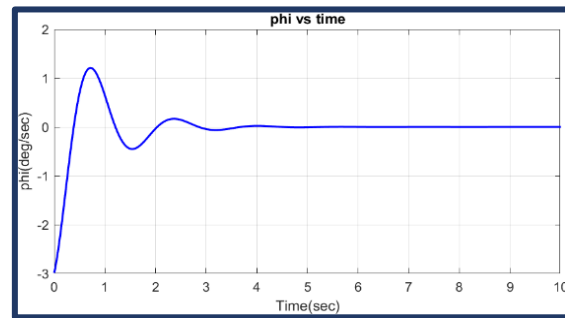


Figure 33: Bank Angle (ϕ) vs Time

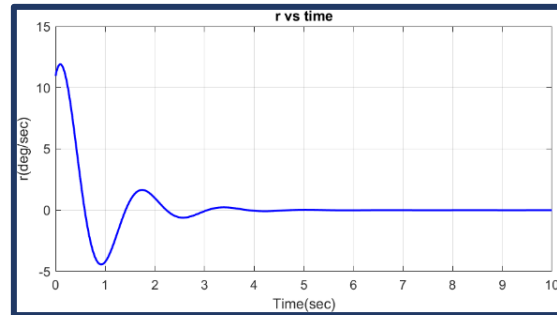


Figure 34: Yaw (r) vs Time

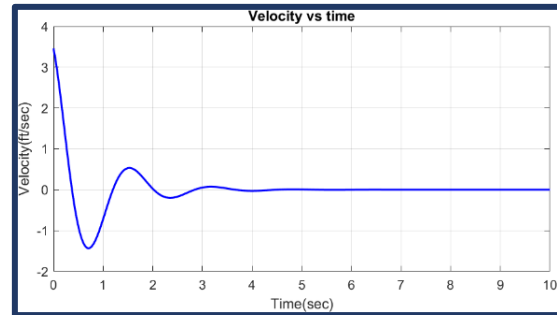


Figure 35: Velocity vs Time

In the graphs, the aircraft's response to the y-direction gusts has been analyzed. The graphs plotted show the lateral stability of the aircraft, wherein the parameters are studied after the disturbance, where the deviations in the plots are dampened over time, indicating to the ability of the aircraft of the aircraft to return to equilibrium through the damped oscillations and maintain stability.

11 PERFORMANCE PREDICTION

11.1 Flight Phase

Table 21: Flight Phase

S. No	Phase	Velocity
1	Take-Off Velocity	39.96 ft/sec
2	Cruise Velocity	55.78 ft/sec
3	Stall Velocity	30.80 ft/sec

$$V_{To} = \sqrt{\frac{2 \times W}{\rho \times S_g \times C_L}}; S_g = \text{Take off distance} = \frac{V_{To}^2}{2 \times g} \times \tan(\alpha); V_{stall} = \sqrt{\frac{2 \times W}{\rho \times S_g \times C_{Lmax}}}$$

11.2 Aerodynamic Performance

$$\text{Lift} = \frac{1}{2} \rho v^2 S C_L \Rightarrow L = 9.7242 \text{ lbs.}; \text{ Drag} = \frac{1}{2} \rho v^2 S C_D \Rightarrow D = 0.7068 \text{ lbs.}$$

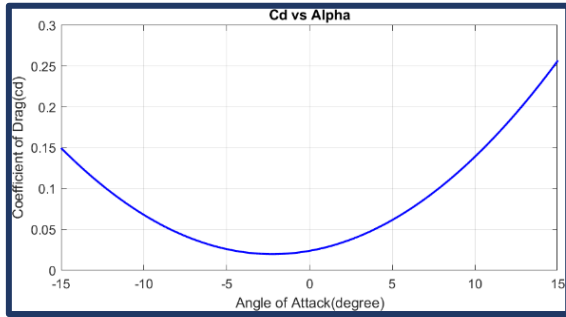


Figure 36: Cd vs Alpha

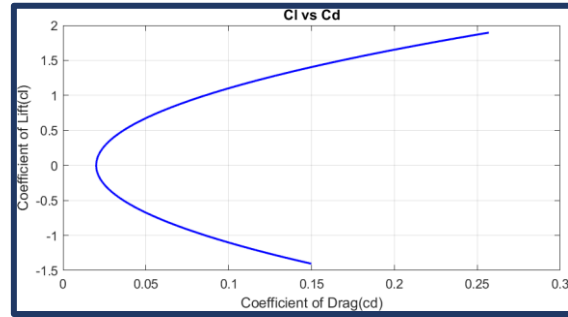


Figure 37: Cl vs Cd

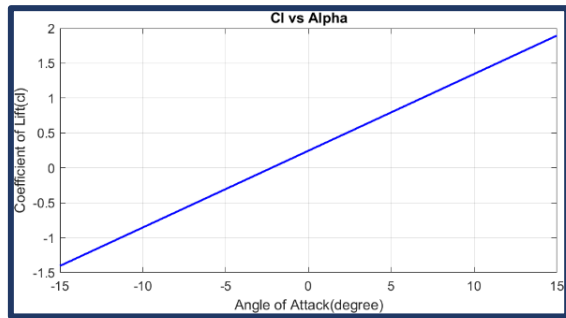


Figure 38: Cl vs Alpha

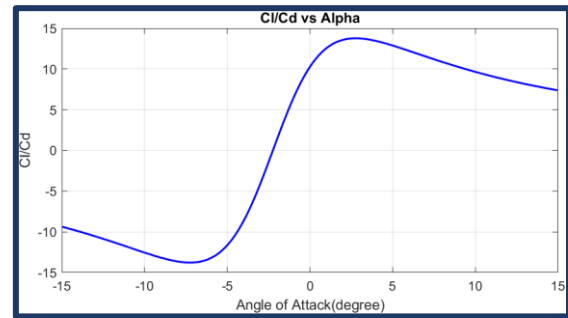


Figure 39: Cl/Cd vs Alpha

11.3 Velocity Behavior

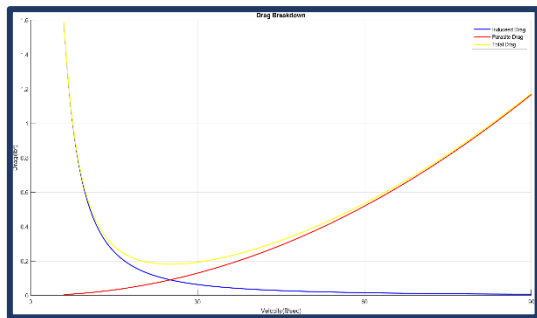


Figure 40: Drag Breakdown

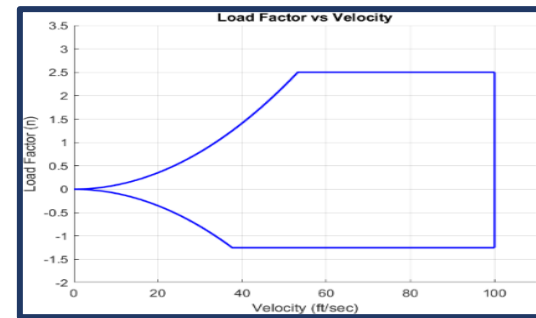


Figure 41: Flight Envelope

- Overall drag (D) = $\frac{1}{2} \rho V^2 S C_D$

$C_D = C_{D0} + K C_L^2$, C_{D0} is parameter of parasite drag and $K C_L^2$ is parameter of induced drag.

$$\text{Parasite Drag} = \frac{1}{2} \rho V^2 S C_{D0}; \text{ Induced Drag} = \frac{1}{\pi e A R} \times \left[\frac{2W}{\rho V^2 S} \right]^2$$

The minimum drag experienced by the aircraft is at 30.80ft/sec velocity. Parasite drag of 0.0200 and an Induced drag of 0.0219 is experienced by the aircraft at cruise speed.

Therefore, Coefficient of drag = $0.0200 + 0.0219 = 0.0419$

- The V-N diagram shows all safe operational regions by using velocity on the horizontal axis and load factor on the vertical axis. The diagram establishes important indicators for positive or negative load factors to guide pilots about aircraft structural limits when conducting maneuvers. The graph stands as a vital safety measure for flying as it directs operators into safe operating zones to stay away from damaging or breaking aircraft structures.

11.4 Range and Endurance

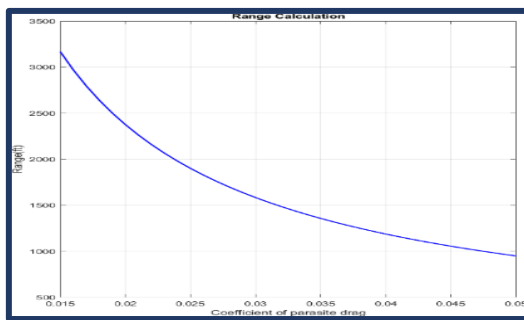


Figure 42: Range

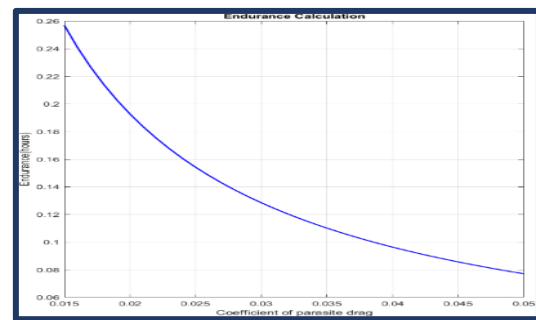


Figure 43: Endurance

- Range(R) - The range is calculated using the endurance and cruise velocity.

$$\text{Range} = \text{Endurance} * \text{Cruise Velocity}$$

Endurance = 4.40 mins; Cruise Velocity = 55.79 ft/sec

$R = 4.40 * 55.79$; $R = 245.432$ ft, The Range in Nautical Miles is 0.0403 NM.

- Endurance(E) – The Endurance is calculated using the battery capacity and the average current drawn by the motor.

$$\text{Endurance} = \frac{\text{Rated Battery Capacity}}{\text{Average Current}} * 60$$

Rated Battery Capacity = 2200mAh; Average current = 30000 mA

$E = \frac{2200}{30000} * 60$; $E = 4.400$ mins, the endurance is 4 min and 40 sec.

12 FABRICATION

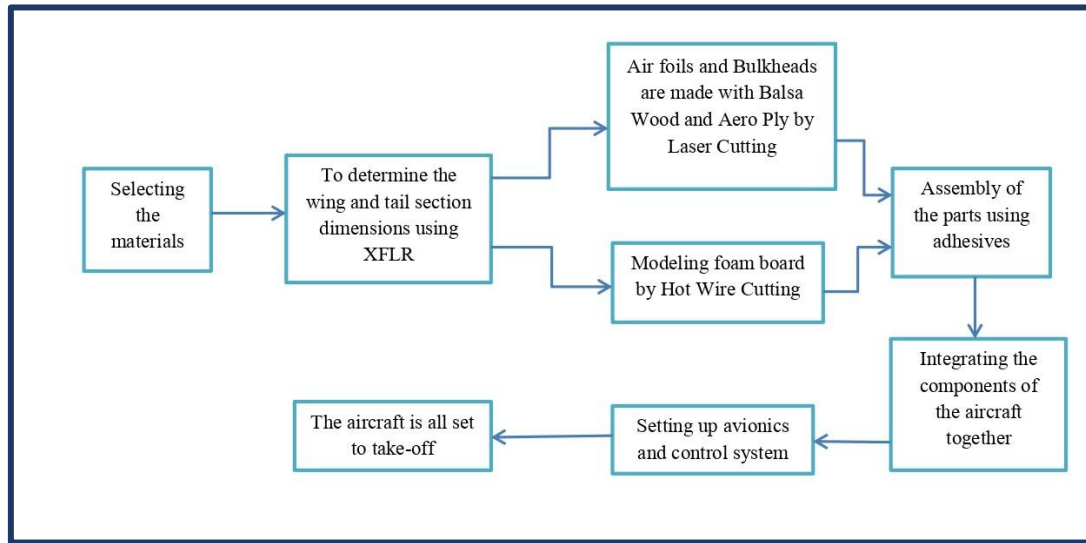


Figure 44: Methodology of Fabrication

12.1 Material Selection

BALSA WOOD: Balsa Wood has been used for the construction of the ribs and the bulkheads of the wing and fuselage respectively as it has an excellent strength to weight ratio.

AEROPLY: We have also used aero ply for the construction of the bulkheads of the fuselage section, since it offers better characteristics than balsa wood.

ALUMINIUM RODS: The aluminum rods were used for the spars in the wing and also for the fuselage rod because these are very light in weight, and are easily available.

FOAM: To make sure that the overall weight of the prototype is considerably lighter, we have used foam to construct the whole body of the prototype.

POLYLACTIC ACID PRO: The polylactic acid pro filament was used to 3D print the structural components for the landing gear and the payload bay.

12.2 Fabrication Techniques

LASER CUTTING: The laser cutting technique has been used to obtain the shape of the ribs and bulkheads for the wings and the fuselage.

HOT WIRE CUTTING: This method was used to shape the foam board in the required shapes with the help of the ribs and bulkheads obtained from laser cutting.

3D PRINTING: The 3D printing for the creation of the components was done, typically used in the landing gear section, where the landing gear has been fixed with the fuselage's bottom, and for the manufacturing of the payload bay.

DRILLING: For the landing gear at the back, drilling machine was used to drill holes on the plywood base, at the which the landing gear rod was fixed with screws.

12.3 Prototype Testing

IARE LAKSHYA MICRO has built and tested a prototype of 1:1 scale with successful flight rounds.

<https://drive.google.com/file/d/1w4Z9XvuRsCHKbpeyWt85AhJOOdwfAwmi/view?usp=sharing>



Figure 45: Prototype Angle 1



Figure 46: Prototype Angle 2

After two (2) flight rounds the landing gear experienced impacts resulting in the bending of the rods and due to consequences, the landing gear had to be removed, but later proved to be still stable for flying

13 SOFTWARES USED

- CATIA V5 and FUSION360 - CATIA V5 and Fusion360 have been used for conceptual design of the aircraft.
- AUTOCAD - AUTOCAD has been used for 2D designing and exporting dxf files.
- ANSYS - The ANSYS software has been used for structural analysis and flow analysis.
- MATLAB - MATLAB has been used for plotting graphs and calculations.
- XFLR5 - XFLR5 has been used for airfoil selection analytics.

14 INNOVATION

- The baffles inside the payload container have been placed at equal intervals, reducing the motion of the fluid resulting in an increased stability.
- The lateral cross section of the fuselage has been designed with an airfoil shape, increasing the overall aerodynamic characteristics and efficiency of the UAV developed.
- The airfoil for the wing has been chosen by the interpolation of four different airfoils resulting in enhancement of the aerodynamics offered by the wing.
- The base connecting the fuselage and the landing gear constitutes the landing gear rod on both sides, allowing for efficient absorption of the forces exerted, resulting in reduced impacts.

15 CONCLUSION

The team IARE Lakshaya Micro has finally constructed the micro class aircraft as per all the requirements and constraints as prescribed by the SAE International Committee. The model that has been developed is capable of carrying a liquid payload of 4.4092 lbs. payload, and is also able to drain the liquid payload within the mission requirement.

16 REFERENCES

- Wing Commander G.M. Jahangir Alam, Dr. Md. Mamun, Dr. Md. Abu Taher Ali, and Dr. A. K. M. Sadrul Islam. **"Development of Design of Airfoil Shaped Fuselage and CFD Investigation of Its Aerodynamic Characteristics."**
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- Ahmed, S., Xin, H., Faheem, M., & Qiu, B. (2022). **Stability analysis of a sprayer UAV with a liquid tank with different outer shapes and inner structures.** *Agriculture*, 12(3), 379.
- Shiva Sharma, Umran Sharieff, Ankit Singh, and R. S. Tarnacha, **"Design and Fabrication of Fixed-Wing UAV for Commercial Monitoring,"** *International Research Journal of Engineering and Technology (IRJET)*, vol. 8, no. 8, pp. 1153-1177, Aug. 2021

APPENDIX: TECH DATA SHEET- AIRCRAFT PERFORMANCE PREDICTION

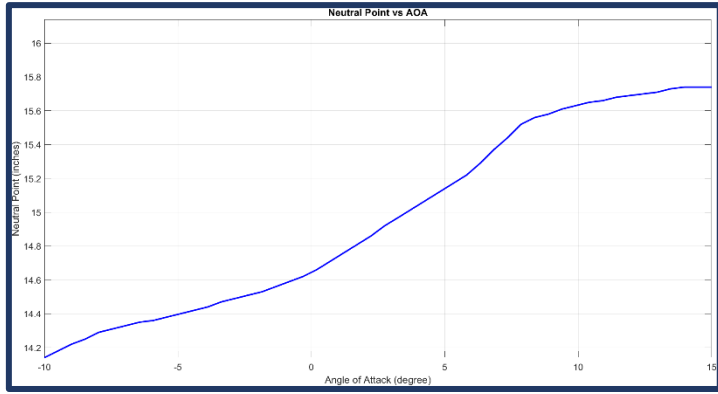


Figure 1: Neutral Point vs AOA

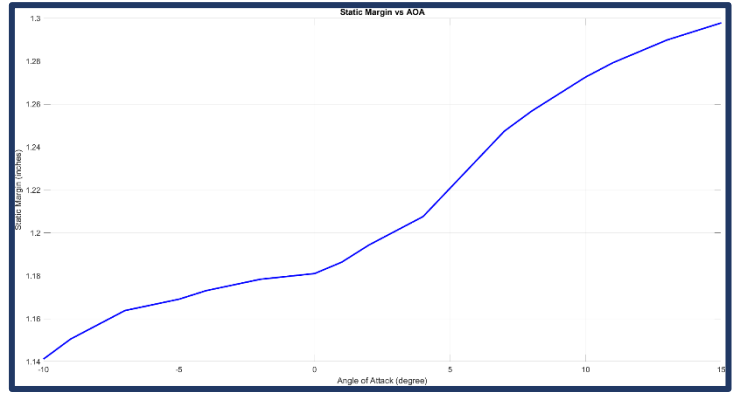


Figure 2: Static Margin vs AOA

$$\text{Neutral Point} = \frac{\bar{C} \times X_{CG} - AC(\bar{C} - h_{CG})}{\bar{C}}$$

$$\text{Static Margin} = \frac{X_{NP} - X_{CG}}{\bar{C}}$$

X_{NP} = NP from Nose Cone

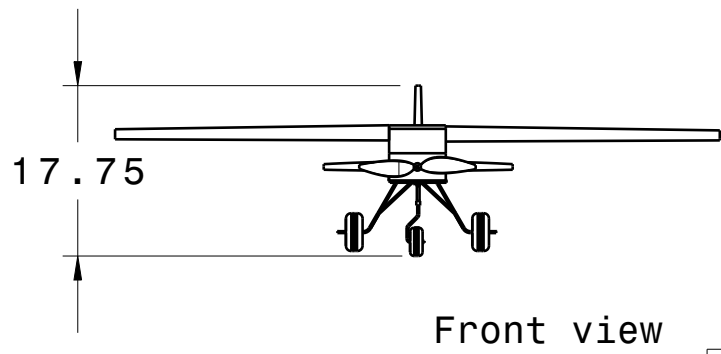
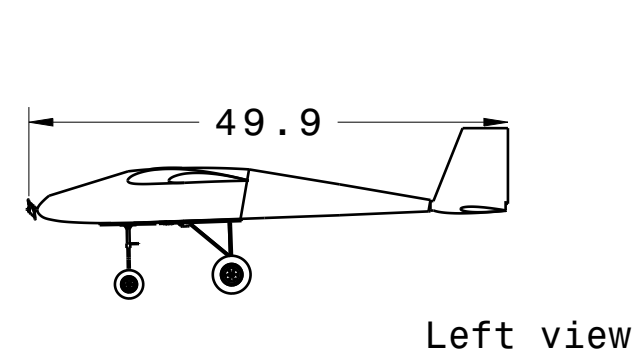
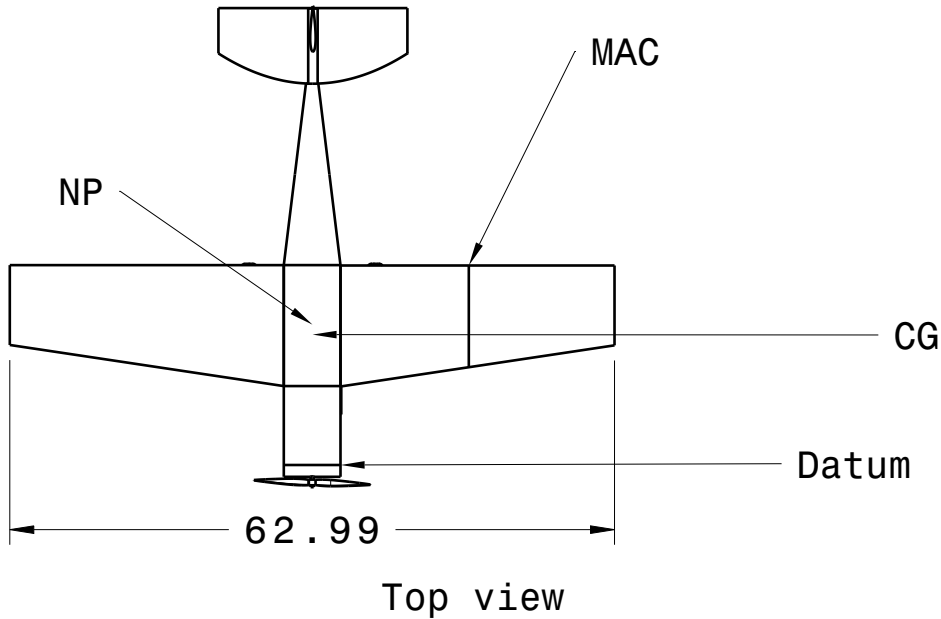
X_{CG} = CG from Nose Cone

The aircraft's longitudinal stability is defined as the tendency of the aircraft to remain stable about its lateral axis, describing the pitching motion caused by the nose up and nose down conditions. This longitudinal stability is determined by the term Static Margin, which is the distance between the center of gravity (CG) and the neutral point (NP). The neutral point (NP) can be defined as a point along the longitudinal axis where the pitching moment does not depend on the angle of attack. The stability of the aircraft changes when the static margin changes, that is, when the CG of the aircraft is shifted forward, the static margin increases resulting in increased stability.

The neutral point (NP) and the center of gravity (CG) at angle of attack 0° , from the datum point, are 374.2 mm and 344.493 mm. The center of gravity (CG) from the neutral point (NP) is 29.7 mm. The static margin in this case is 11.051 %. As we have a liquid payload, the movement of the water accounts in the back-and-forth movement of the center of gravity (CG), and the neutral point (NP) also keeps shifting, resulting in change in stability. The graphs plotted show that the static margin increases as the angle of attack of the aircraft increases, which leads to the change in the pitching moments, thus influencing the overall stability characteristics of the aircraft.

TEAM NUMBER	21431953
TEAM NAME	IARE LAKSHYA Micro
College	Institute of Aeronautical Engineering

Wing span	62.99213 in
Wing area	657.20131 sq.inches
Taper ratio	0.65625
Aspect ratio	6.0377
Empty Weight	4.9399 lb
Maximum Take-off weight	9.6136 lb
Static Margin	11.051 %



Components	Mass (lb)	Weight (lbf)	Xcg (in)	Moment at cg (lbf-in)
Propeller	0.060	1.9151	-2.126	-4.07157
Reciever	0.033	1.0640	3.543	3.76997
ESC	0.179	5.7454	1.235	7.09810
Power Limiter	0.044	1.4186	2.953	4.18886
Battery -2200mAh	0.386	12.4130	1.235	15.33540
Motor	0.317	10.2141	-0.886	-9.04793
Fuselage	0.415	13.3562	19.291	257.65941
Wing	1.725	55.5087	13.780	764.88281
Horizontal Stabilizer	0.178	5.7300	42.992	246.34370
Vertical Stabilizer	0.089	2.8622	43.110	123.38792
Payload	4.674	150.374	13.552	2037.92780

Servo Position	Mass (lbs)	Manufacturer	Model Number	Torque(ounce-inch)
Starboard Aileron	0.0024720	Tower Pro	SG90	24.9972958
Port Aileron	0.0024720	Tower Pro	SG90	24.9972958
Elevator servo	0.0024720	Tower Pro	SG90	24.9972958
Fin servo	0.0024720	Tower Pro	SG90	24.9972958
Landing Gear Servo	0.0083150	EMAX	ES3001	44.4396369

Battery capacity	2200 mAh
Motor make and Model	T- Motors, AS2820 Long Shaft
Motor KV	1250KV
Propeller Manufacture	APC Propellers
Propeller diameter	12 inches
Propeller pitch	6 inches

Scale 1 : 20	All dimenisons are in Inches	Sheet 1/1
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